

ENTRY 022 Validation Harness Correction — August 6, 2026 — Status: COMPLETE

Entry 022 — v3's First Honest Validation: Two Failures Pointing Opposite Ways

The Cairo Harness Was Silencing Half Its Noise. Replacing It Immediately Exposed a Missing Decoherence Term.

1. What Was Done

Entry 021 closed by naming `bell_scattered_0_26` — a 14-point discrepancy — as the largest remaining error in the emulator and the next priority. This entry set out to diagnose that single circuit. It found instead that the validation harness itself was defective, that no circuit-level accuracy figure from Entry 014 onward was measurable, and, once a sound harness was built, that v3 is missing a physical term entirely.

2. First Defect — the Basis-Gate Mismatch

FakeCairoV2 is a **mixed-basis snapshot**: 12 of its directed edges are calibrated as `cx`, the other 14 as `ecr`. The harness transpiled with `basis_gates = backend.operation_names`, handing the transpiler the union of both. The transpiler selects one two-qubit gate for the whole circuit — `cx` — and emits it on every edge, including the fourteen that only carry `ecr` calibration.

Qiskit's noise model keys its quantum errors by *(gate name, qubit pair)*. For those gates it finds no matching entry and applies **nothing at all**. The gate executes perfectly, silently, for free.

Circuit	2q gates	noisy	SILENT	% free
<code>ghz_local_0_1_2</code>	2	0	2	100%
<code>ghz_scattered_0_26_22</code>	35	20	15	43%
<code>bell_adjacent_24_25</code>	1	1	0	0%
<code>bell_scattered_0_26</code>	34	13	21	62%
TOTAL	72	34	38	53%

Over half of every two-qubit gate in the suite ran without error. `ghz_local_0_1_2` — the circuit that had looked most accurate throughout — ran with zero noise on both of its gates.

3. Second Defect — Calibration Coverage

Independently of the basis problem, only **12 of Cairo's 28 topology edges** carry real calibrated error values — 42.9%. The remaining 16 fell back to v3's hardcoded 1% constant. On that chip, most of what the model reported was not a measurement of anything.

4. A Sound Harness — FakeKyiv

FakeKyiv has neither defect: uniformly `ecr`, noise-model coverage on all 144 edges, and 139/144 (96.5%) real calibration coverage. The replacement harness, `validate_v3_kyiv.py`, audits its own noise coverage on every run and declares its results void if any gate runs silent.

One further trap was found in the process. Aer's automatic simulation method returns `success=True` and `status=COMPLETED` for the deeper routed circuits, then raises on `get_counts()`. A harness trusting it would silently lose precisely its hardest cases. `matrix_product_state` is mandatory here, not preferred.

5. Results — All Gates Verified Noisy

Circuit	hops	2q gates	Aer reference	v3	gap
<code>bell_adjacent_77_78</code>	1	1	96.07%	99.25%	3.18
<code>bell_near_77_82</code>	5	13	48.74%	92.89%	44.15
<code>bell_mid_77_100</code>	6	16	83.69%	90.27%	6.58

Circuit	hops	2q gates	Aer reference	v3	gap
bell_far_0_126	26	76	69.90%	60.51%	9.39
ghz_local_77_78_79	2	2	88.53%	98.67%	10.14

Mean absolute gap 14.69 points. The mean is misleading, and that is the finding. (ghz_scattered_0_63_126 was killed mid-run — two 26-hop routes exceeded MPS memory — and is excluded.)

6. The Anomaly — 13 Gates Worse Than 76

Thirteen gates produce 48.74%. Seventy-six gates produce 69.90%. Six times the gate count, a better outcome. Circuit depth does not explain this chip's behaviour.

The 77→82 route passes through **qubit 80: T2 = 8.51 μs, T1 = 5.02 μs** — effectively dead against a chip median in the hundreds. With a 560 ns ecr, that is roughly 6.6% dephasing per gate, and the route crosses it six times.

Route	worst T2 on path	2q gates	Aer reference
77 → 82	8.5 μs (qubit 80)	13	48.74%
77 → 100	32.7 μs	16	83.69%
0 → 126	healthy throughout	76	69.90%

Compounding the failure: edges (79,80) and (80,81) are among the five Kyiv edges with **no calibration data**, so v3 applied its 1% fallback guess at exactly the point where reality is catastrophic. Two independent weaknesses landed on the same two edges.

7. Diagnosis — No Decoherence Term

v3 models two-qubit *gate error* only. It has no decoherence term at all. The consequences run in opposite directions:

- **Short routes through low-T2 qubits are catastrophically under-penalized** — 44 points on 77→82.
- **Long routes through healthy qubits are over-penalized** by SWAP compounding — 9 points on 0→126, with v3 pessimistic.

Because the two errors have opposite sign, any single mean accuracy figure conceals both. This is Entry 021's finding resurfacing one level up: there, the domination factor $D = \text{gate error} \div (\text{gate duration} \div T2)$ was introduced as a *screening* criterion for which edges were fit to measure. Entry 022 shows the same quantity belongs *inside the model*. Every edge on the 77→82 route through qubit 80 sits at $D \approx 0.15$ — the decoherence-dominated regime Entry 021 identified.

8. A Naive Fix, and Why It Was Rejected

The obvious correction is to charge $\text{gate duration} \div T2$ per gate alongside the gate-error term. Tested on the real transpiled routes:

Circuit	Aer reference	v3 gate-only	v3 + decoherence
bell_near_77_82	48.74%	93.10% (gap 44.36)	58.83% (gap 10.09)
bell_far_0_126	69.90%	64.06% (gap 5.84)	30.67% (gap 39.23)
MEAN ABSOLUTE GAP	—	14.10	13.85

It repairs the short bad route and destroys the long good one; the mean barely moves. The reason is physical. In a SWAP chain **the state moves**, so each qubit holds it only briefly. Charging every gate the full T2 cost of both its qubits double-counts heavily across 76 gates spread over 26 healthy qubits, while being roughly right for 6 gates parked on one dead qubit.

Decoherence must be **time-integrated along the carrier path** — summed, over each qubit the state actually occupies, of time spent there $\div$ that qubit's T2 — not applied per gate to both members of every pair. That is the v4 design.

9. What This Means for QuantumBridge

No v3 accuracy figure predates this entry. Every circuit-level number from Entry 014 onward was measured against a reference silencing half its noise. The forgiveness work of Entries 017–021 is unaffected — those used single-edge circuits with no routing, on properly covered edges — but the emulator validation figures must be treated as withdrawn.

FakeCairoV2 is retired for validation. Mixed basis and 42.9% calibration coverage make it unusable for this purpose. Kyiv or Sherbrooke only.

The fallback constant is actively dangerous. It fires on uncalibrated edges, and uncalibrated edges correlate with broken qubits — precisely where a wrong guess costs most. v4 should refuse to predict on an uncalibrated edge rather than quietly assume 1%.

Routing is error-blind. v3's BFS minimizes hop count while ignoring both edge error and T2. The 77→82 route is the shortest available and also among the worst.

Entry 022 — Conclusion

An investigation into one anomalous circuit found the measuring instrument broken. The Cairo validation harness emitted cx gates on ecr-calibrated edges, where the noise model applied no error whatsoever — 53% of all two-qubit gates in the suite, and 100% of them in the circuit that had appeared most accurate. Rebuilt on FakeKyiv with per-run noise-coverage auditing, the first trustworthy measurement shows v3 failing in two opposite directions at once: under-penalizing short routes through low-T2 qubits by 44 points, and over-penalizing long routes through healthy ones by 9. The cause is a missing decoherence term. A naive per-gate correction was tested and rejected because it double-counts along a SWAP chain, which localizes the requirement: decoherence must be integrated along the path the quantum state actually travels. The headline number from this entry is not an accuracy figure — it is that the project did not previously have one.

10. Next Step — Entry 023

Two changes, in order. First, remove the 1% fallback so v3 declines to predict on uncalibrated edges instead of guessing. Second, implement v4's time-integrated decoherence term and re-validate against the Kyiv reference established here. A third question follows from Section 9: whether routing should be made error-aware and T2-aware, since the shortest path is demonstrably not the cheapest one.

Research Log — Index Addendum (Entry 022)

Entry	Title	Date	Status
Entry 022	Validation Harness Correction — Basis Mismatch and the Missing Decoherence Term	Aug 6, 2026	Complete

Entries 014–016 and the circuit-level validation figures of Entries 019 and 021 are marked **measurement void** by this entry: their references silenced 53% of two-qubit noise. Their methodology and reasoning stand; only the accuracy numbers are withdrawn.